BANK CUSTOMER CHURN ANALYSIS

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Business Problem

The bank seeks to analyze customer churn data to identify which customer segments exhibit significant churn rates and when these churn events most frequently occur. The objective is to obtain sufficient insights to develop effective retention strategies and, where appropriate, adjust lead targeting toward profiles that demonstrate lower churn tendencies.

To achieve this, the management team has tasked the data analyst with examining customer churn records and extracting actionable insights. The analysis will focus on profiling customers with the highest churn rates as well as those with the lowest churn rates, providing a comprehensive view to guide strategic decision-making.

Key Business Questions

Customer Profile & Demographics

- Which demographic segments show the highest churn rates?
- Does tenure correlate with churn likelihood?
- Are younger or older customers more likely to exit, and how does this vary by geography?

Financial & Product Behavior

- How does credit score influence churn? Are lower-score customers more likely to leave?
- Is churn higher among customers with low balances or fewer products?
- Do customers with multiple products exhibit lower churn?
- How does card type relate to churn behavior?

Engagement & Satisfaction

- Are active members less likely to churn compared to inactive ones?
- How strongly does satisfaction score predict churn?
- Do customers who file complaints have a higher probability of exiting?
- Is there a threshold of points earned that correlates with loyalty or retention?

Income & Value Segmentation

- Does estimated salary influence churn? Are high-income customers more stable or more mobile?
- Which segments represent the highest value customers who are at risk of churn?

Timing & Patterns

- At what tenure milestones does churn spike?

Strategic Targeting

- Which high-risk segments should be prioritized for retention strategies?

Dataset Overview

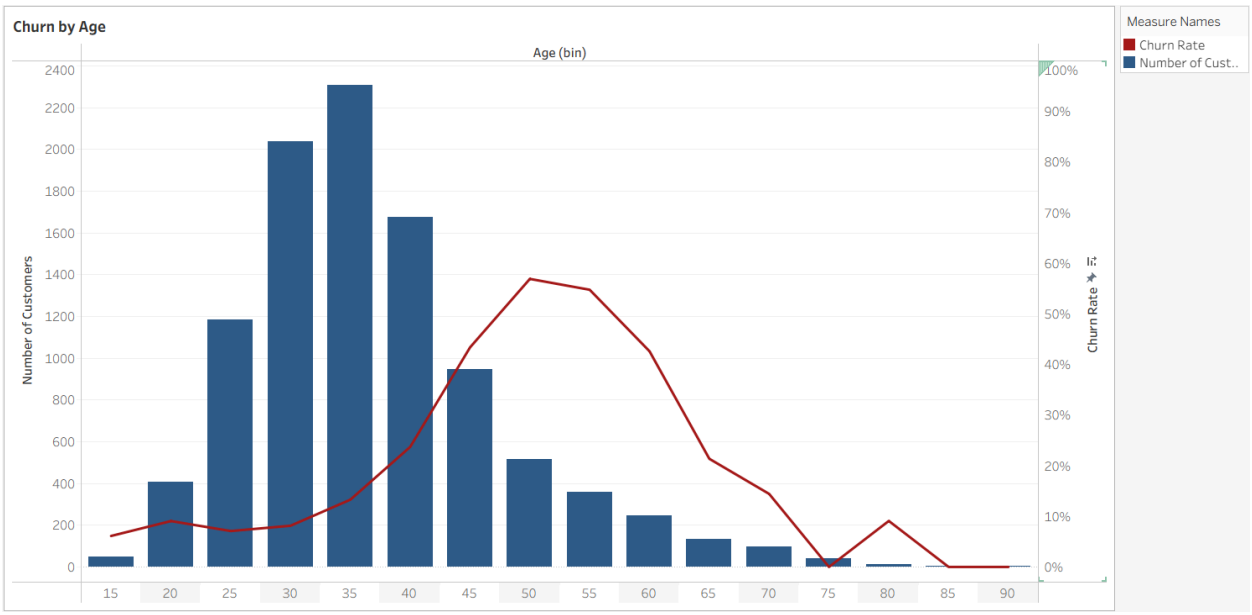
The dataset includes the following variables:

- Customer ID
- Surname
- Credit Score
- Geography (country)
- Gender
- Age
- Tenure (years)
- Balance: money in account
- Number of products
- Has Credit Card (yes/no)
- Is Active Member (yes/no)
- Estimated Salary
- Exited (yes/no)
- Complain (yes/no)
- Satisfaction Score (1 to 5)
- Card Type
- Points Earned

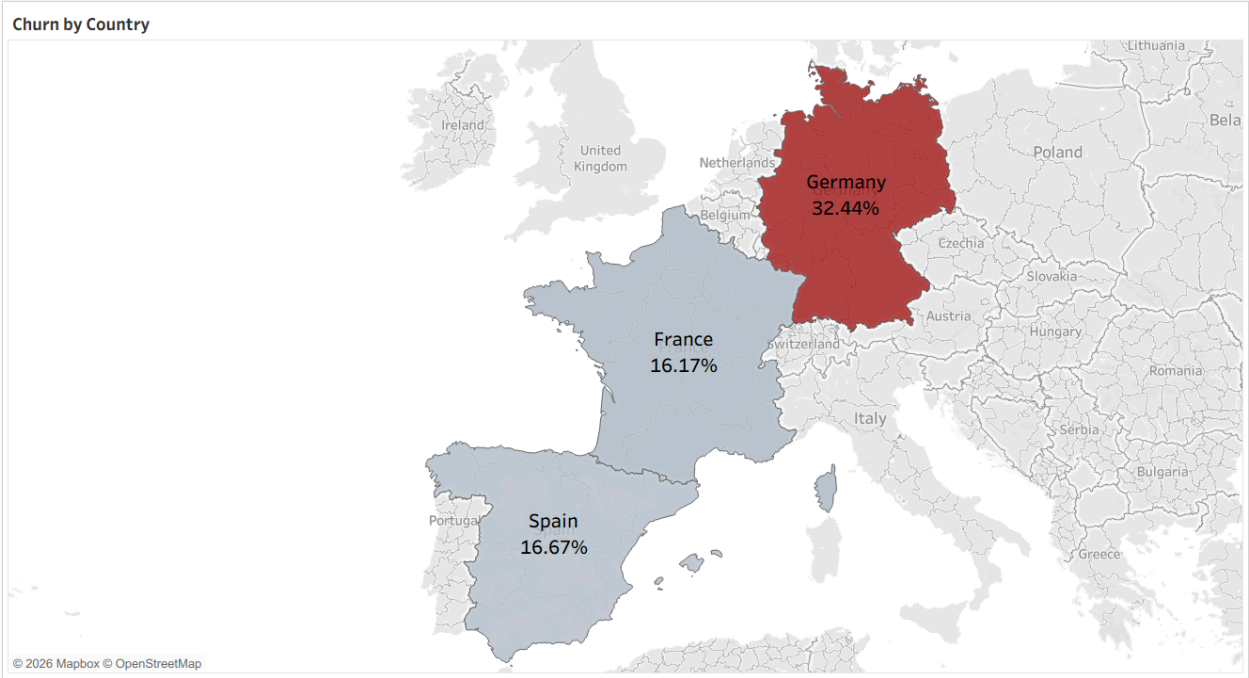
ANALYSIS AND FINDINGS

The dataset includes 10,000 customers, of whom 2,038 have discontinued the bank's services, resulting in an overall churn rate of 20.38%. The average customer age is 39, with ages ranging from 18 to 92. The majority (82%) fall within the 25–50 age range.

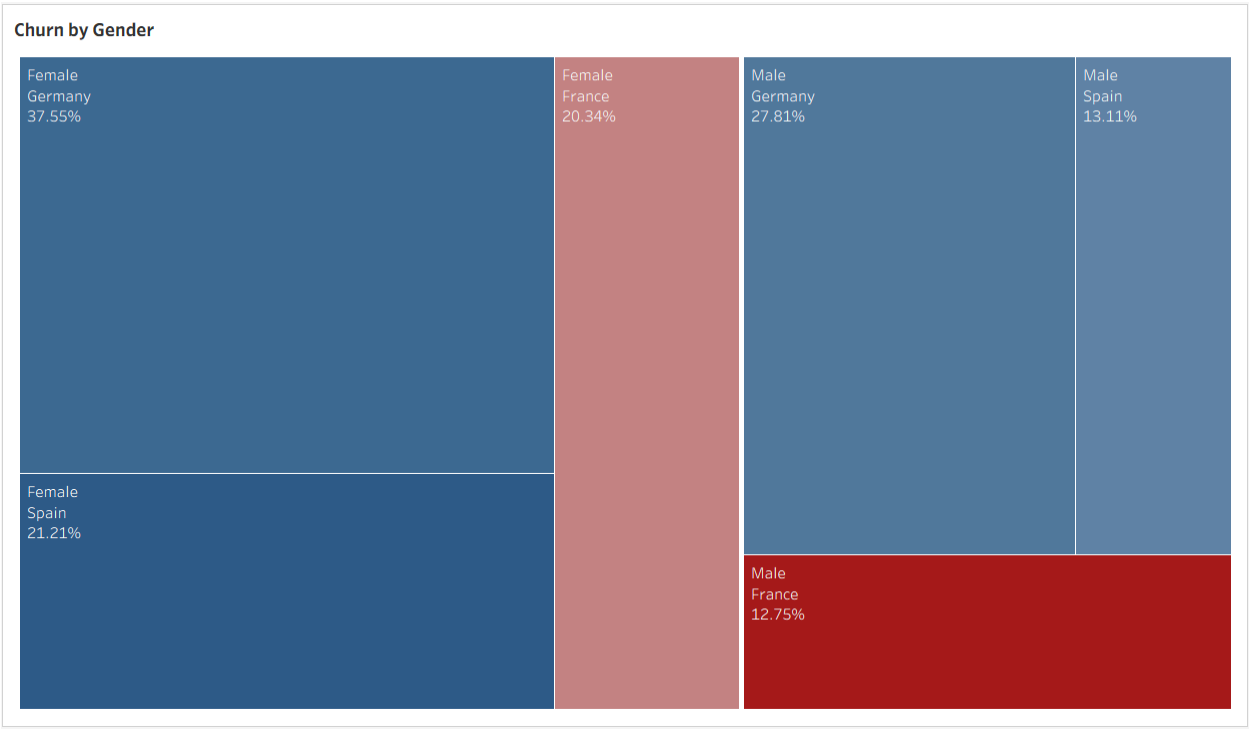
The chart shows that the churn rate begins to rise in the 35–40 age group and becomes significantly higher in the 50–55 age group, after which it drops sharply. This pattern may indicate that, beyond a certain age, a subset of customers no longer finds the bank's services fully aligned with their needs and chooses to exit. Past this age threshold, the customer base becomes significantly smaller; however, the churn rate decreases sharply, suggesting that the remaining customers represent a more stable segment. This stability may reflect better service alignment or higher switching costs. This pattern appears consistently across the three countries in which the bank operates.



The bank serves customers across three countries, with half of them located in France (50%) and the remainder almost evenly distributed between Spain (25%) and Germany (25%). According to the map chart, Germany exhibits the highest churn rate, double that of France and Spain.

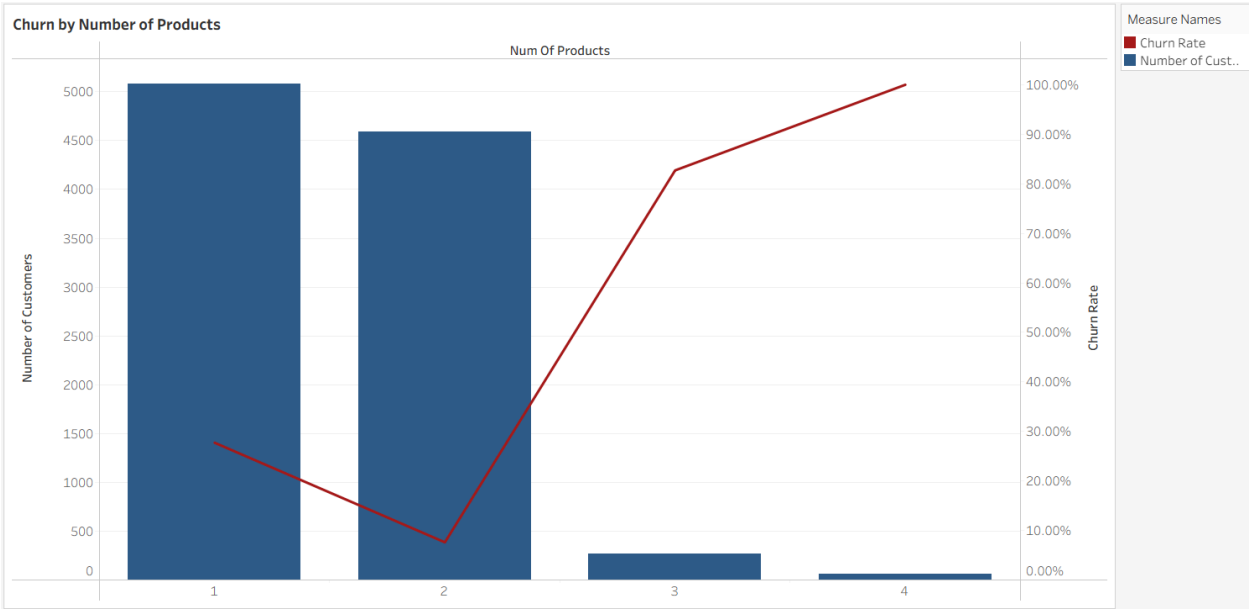


Regarding gender, women are more likely to unsubscribe from the bank's services, with a churn rate approximately 10 percentage points higher than that of men.

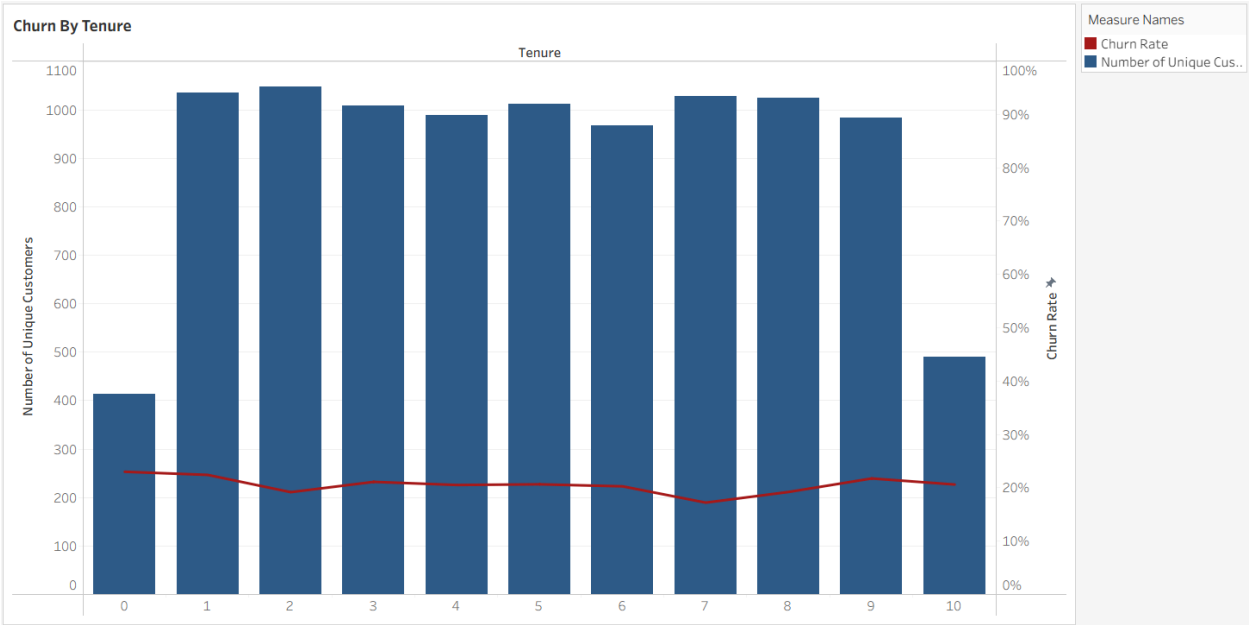


Customers subscribe from 1 to a total of 4 different products, and customers subscribe to an average of 1.53 of them. As shown in the chart, customers with one product have a churn rate of 28% and represent the largest customer segment. Among customers with two products, the number of subscribers decreases slightly, but they are significantly less likely to churn.

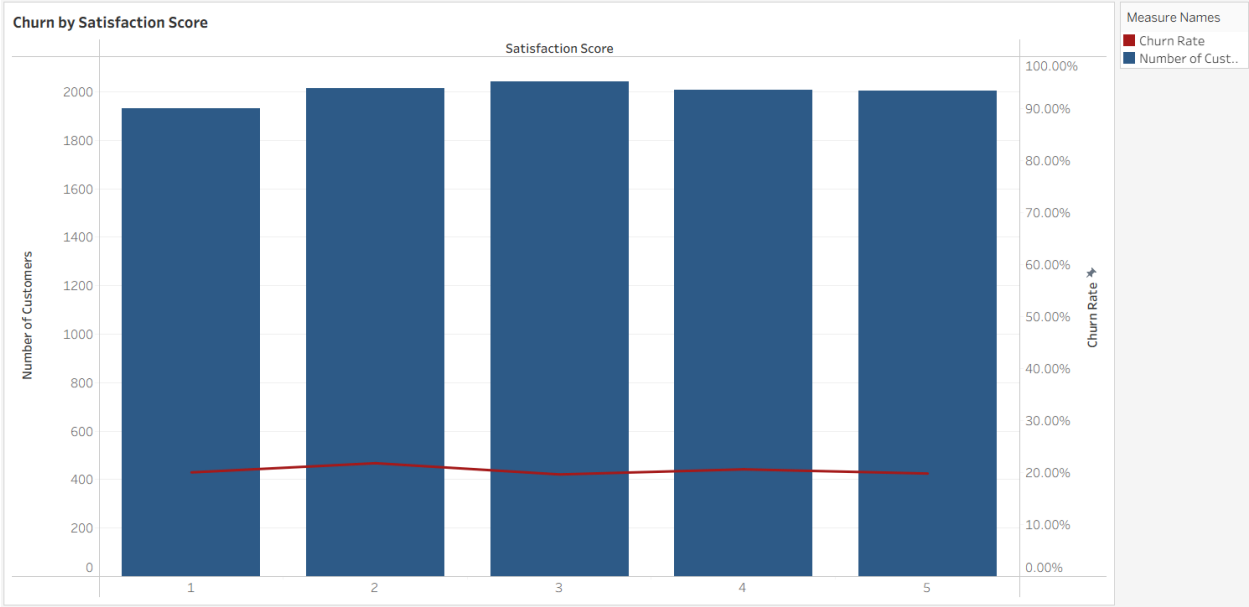
The situation changes completely for customers with three or four products: their churn rates are much higher, and the number of active customers in these groups is very small.



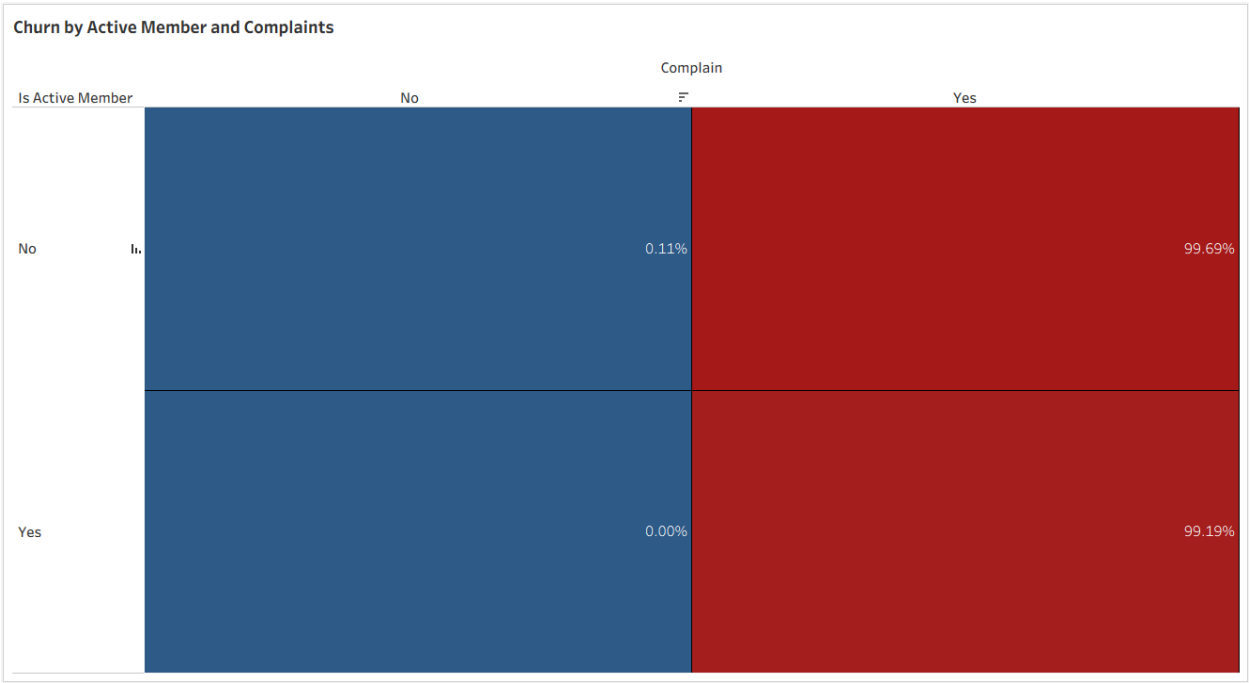
Tenure shows to not be an influence in churn rate, as the trend shows no meaningful variations. Although churn rate remains consistent across tenure groups, customer counts are notably lower at both ends of the tenure spectrum. This may reflect acquisition slowdowns, historical growth patterns, or natural attrition, rather than tenure-driven churn behavior.



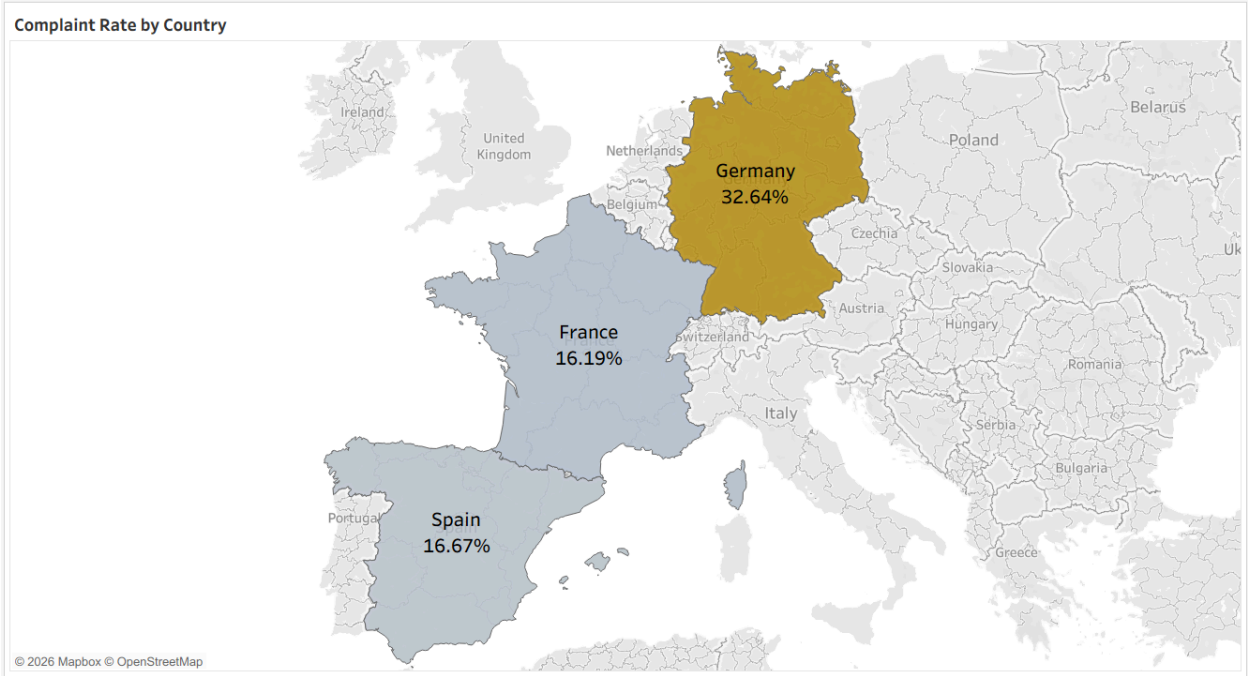
No significant patterns were observed in churn behavior across satisfaction scores, even when segmented by gender and geography.



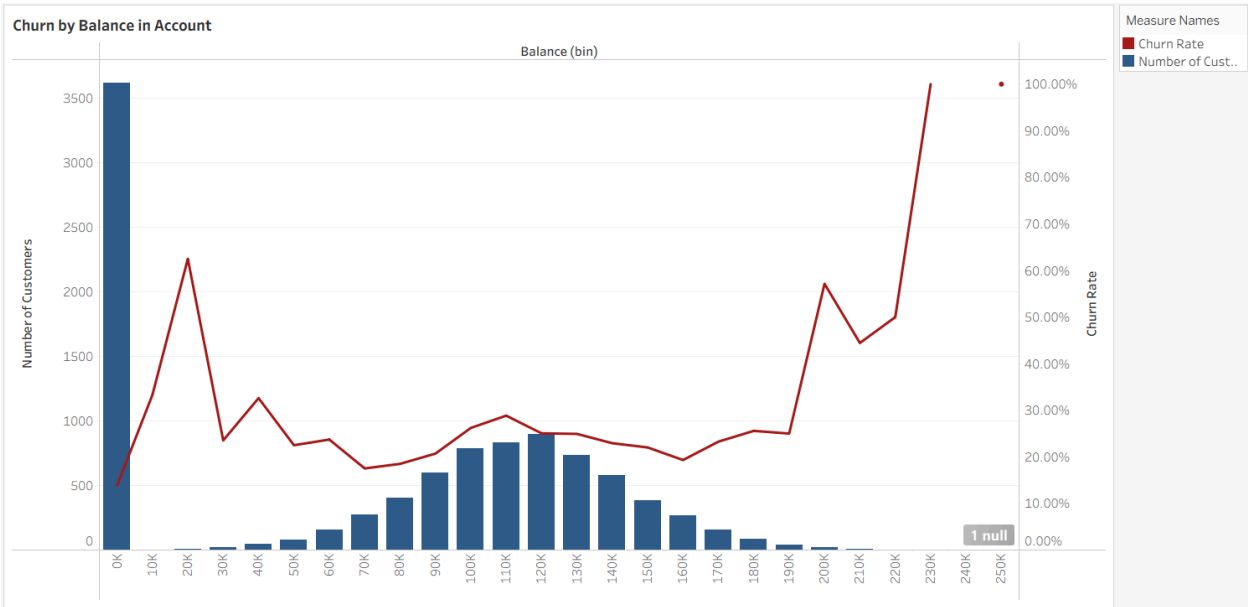
Churn rate across membership status and complaints reveals a striking pattern. Customers who have complained, whether active or not, show an almost 100% churn rate, while those who have not complained exhibit near-zero churn, regardless of activeness. This may indicate that certain issues raised by customers are severe enough to drive cancellation, suggesting they act as deal-breakers for continued service use.



Customer complaints emerge as the dominant churn driver across all geographies. At a country level, churn rates closely mirror complaint rates, and at an individual level, customers who complain churn almost universally. This indicates that geography influences churn primarily through differences in complaint behavior rather than through demographic or financial characteristics alone.



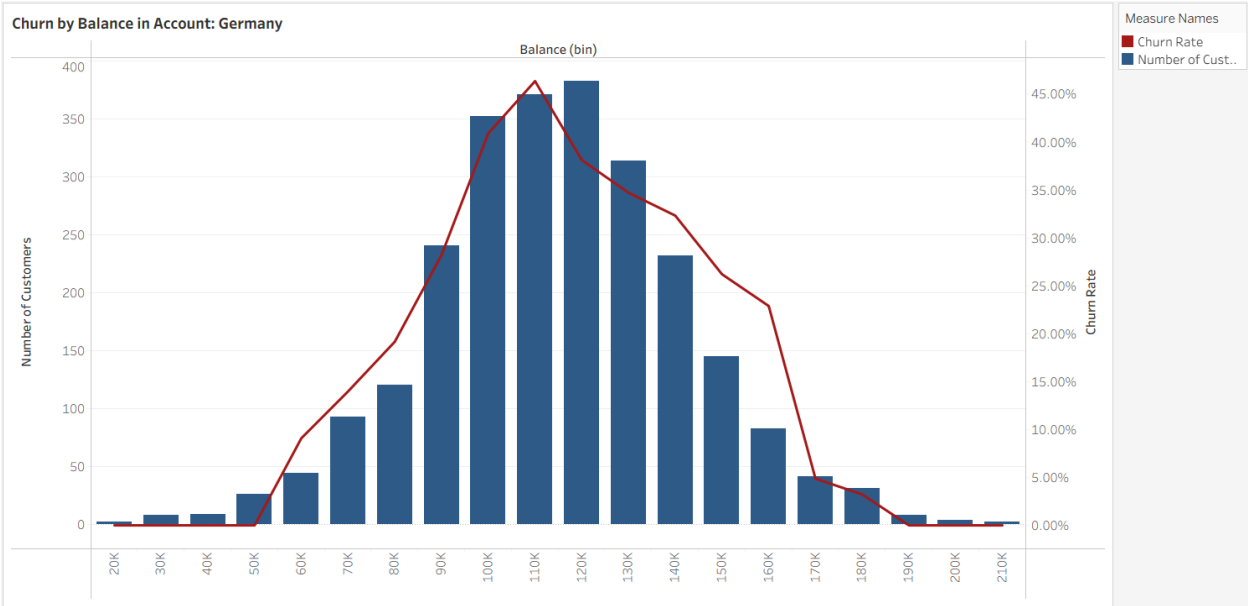
Excluding the segment of customers with 0K to 10K of balance in account, churn rate remains relatively stable across the balance brackets where most customers are concentrated. It only spikes at the extremes, among customers with balances around 20K–40K, and again among those with very high balances above 200K, though these outer groups include very few customers. Overall, account balance does not show a consistent relationship with churn.



Of all customers in the dataset, 48% are inactive, and 36% of all customers (active and inactive) have a balance of $0. Excluding customers who have churned, since they are no longer clients, a total of 14% of all customers have an account but do not use it. Those who have churned account for an additional 3% of the total.

Among these customers who do not use their accounts, 66% are from France and 33% are from Spain. Germany has none, and it also has no customers with a $0 balance or with very little money.

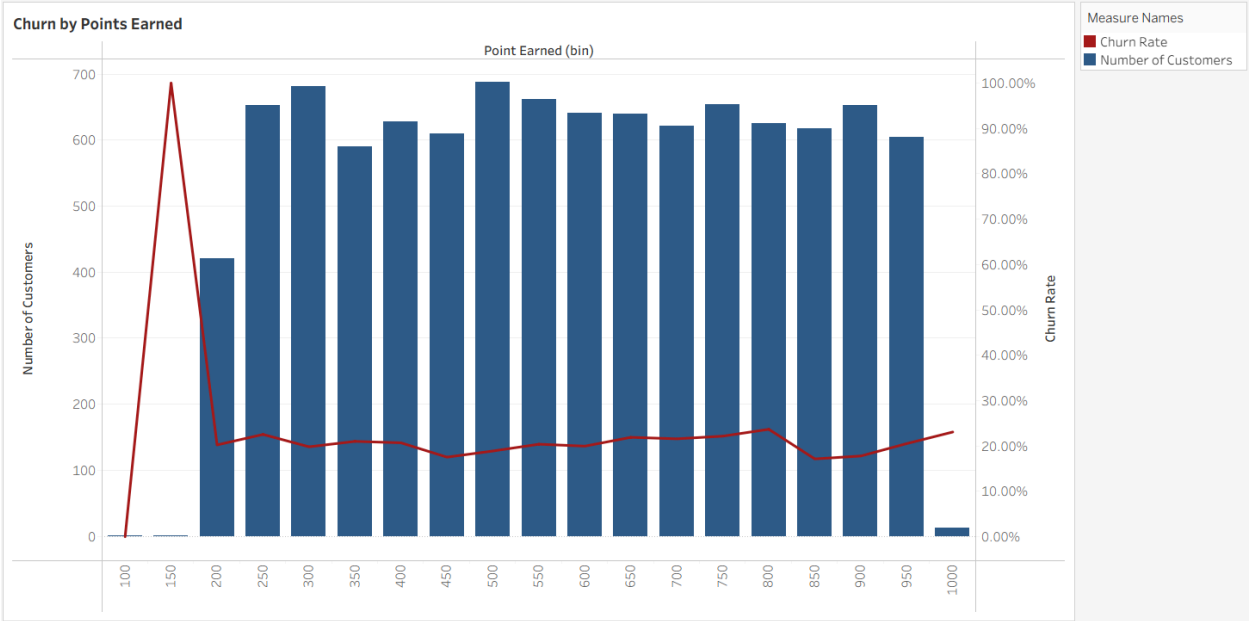
At an aggregate level, account balance does not appear to be a strong churn driver. However, when segmented by geography, a clear interaction emerges. In France and Spain, churn remains relatively stable across balance levels, suggesting balance has little explanatory power. In contrast, Germany shows a structured churn pattern by balance, with significantly higher churn among mid-to-high balance customers. This indicates that geography moderates the relationship between balance and churn, and that high-value German customers represent a critical retention risk.



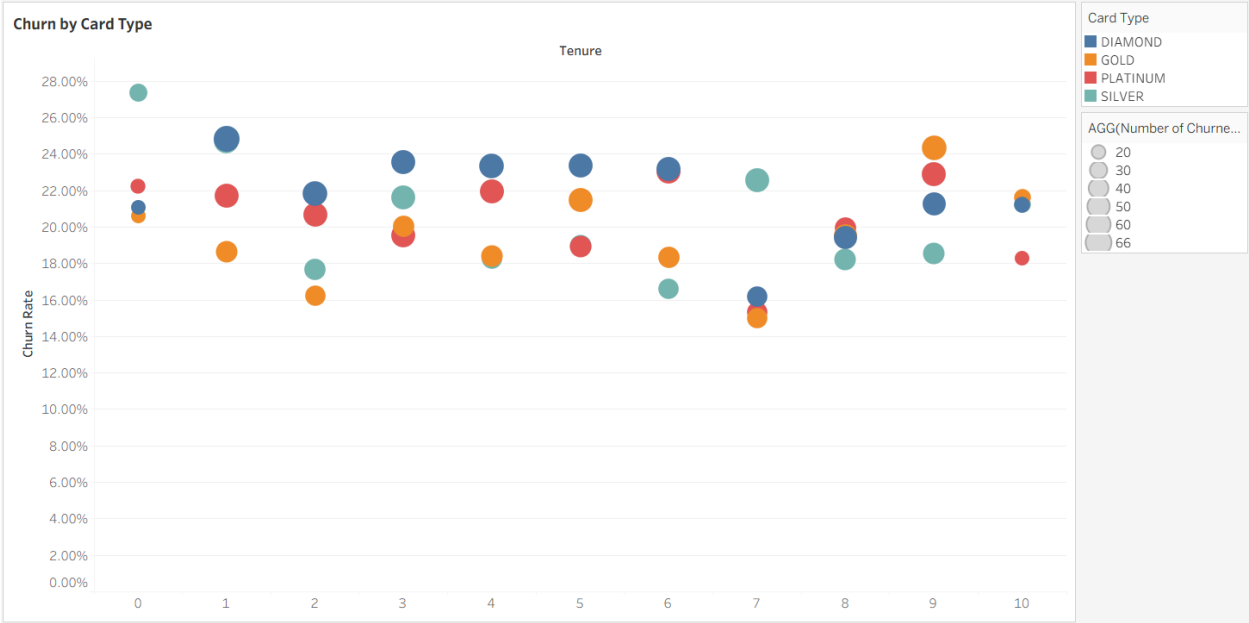
An analysis of churn rate across different levels of points earned shows no meaningful relationship between these two variables. When examining the full dataset, churn rates remain stable across all point ranges, indicating that the number of points earned does not influence the likelihood of a customer churning.

When the data is segmented by country and gender, the churn rate lines become noticeably noisy and irregular. These fluctuations do not represent a real trend.

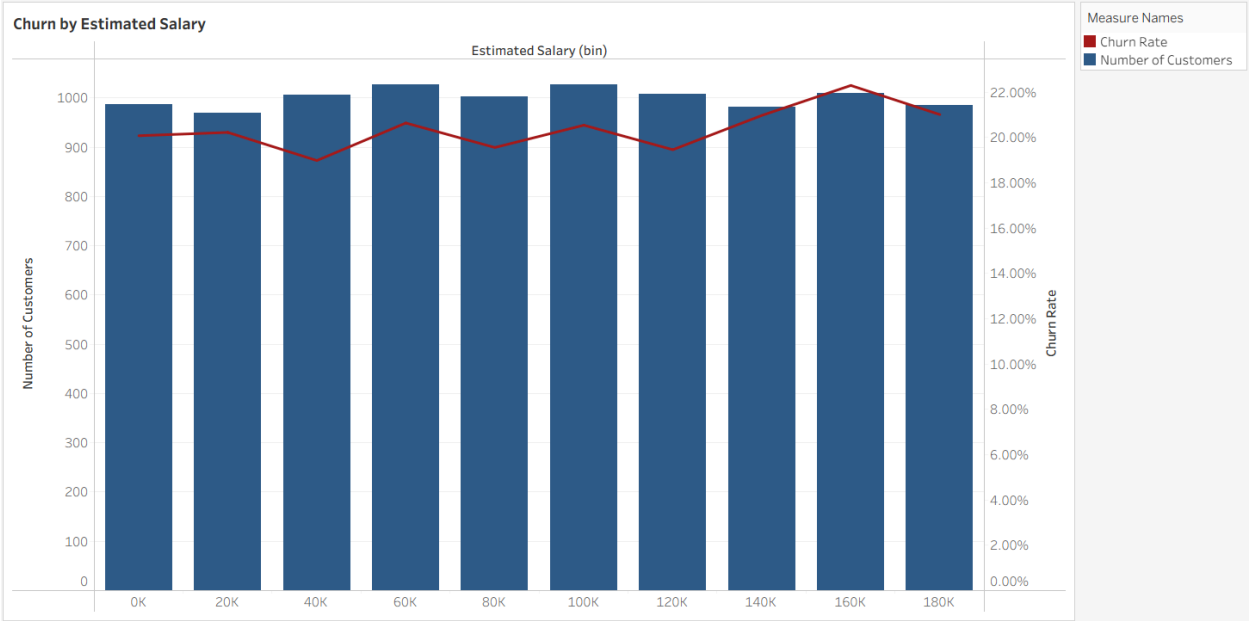
Overall, both the stable trend in the aggregated data and the noise observed in the segmented charts consistently indicate that points earned are not a significant driver of customer churn.



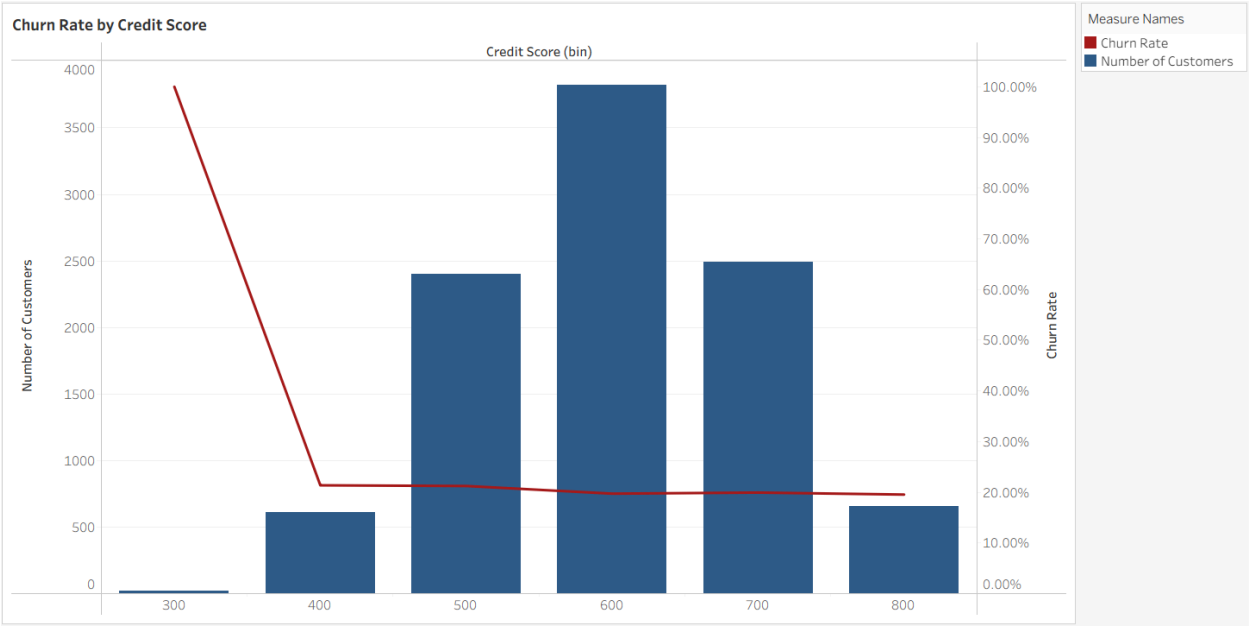
Card type does not appear to have a meaningful impact on churn rate. Across all tenure levels, the churn rates for Diamond, Gold, Platinum, and Silver are all relatively close, with no card type consistently performing better or worse.



Salary is also not an indicator of churn rate, as the trend line remains largely horizontal, and the number of customers in each salary group does not vary significantly.



The vast majority of customers have a credit score between 500 and 800. However, this variable does not appear to be a meaningful indicator or driver of churn, as the churn rate remains stable across most credit score brackets. The only exception is the 300-score bracket, which shows higher variability but represents a very small number of customers.



Data limitations

The dataset does not include timestamped data such as customer onboarding dates, churn dates, or complaint dates. Although customer tenure (in years) is available, it does not enable time-based analyses. As a result, it is not possible to evaluate how churn rates have

evolved over time, conduct cohort analyses, or assess the sequence and timing between complaints and customer churn.

While the dataset indicates whether a customer has submitted a complaint, it does not include any reason codes. This prevents sequence analysis to identify the specific drivers of dissatisfaction behind the complaints.

The dataset does not specify the total number of products offered by the bank. While customers in the data are subscribed to a maximum of four products simultaneously, this does not necessarily imply that the bank offers only four products overall. As a result, conclusions regarding churn among customers with three or four products should be interpreted with caution, particularly given the small size of these segments.

MARKETING RECOMMENDATIONS

Complaint Handling as the Primary Retention Strategy

The bank should prioritize complaint resolution and root-cause analysis. Retention efforts should focus on preventing complaints where possible and resolving them promptly when they occur.

Churn in this dataset is almost entirely complaint-driven, with complaints acting as a leading indicator and likely trigger for customer exit. Among customers who submitted a complaint, 99.5% subsequently exited the bank's services. While the majority of customers do not complain (overall churn rate of 20.39%), those who do almost invariably churn, highlighting the critical importance of effective complaint management.

Geographic Market Nuances

German customers submit complaints at approximately twice the rate observed in France and Spain, which show similarly lower complaint levels. This indicates that Germany represents an operational risk rather than an inherent customer loyalty issue.

France accounts for approximately 50% of the customer base, while Germany and Spain each account for 25%. This distribution, combined with France's lower churn and complaint rates, suggests higher customer engagement and positions France as the most attractive and stable market. Germany, in contrast, requires targeted operational improvements to address the root causes of customer dissatisfaction.

Product Offering Efficiency

Customers typically subscribe to one or two products, with very few holding three or more. However, churn rates increase sharply among customers with multiple products. Given that complaints appear to reflect critical service failures, the bank should focus on improving its ability to deliver and manage multiple products simultaneously. Enhancing service efficiency for multi-product customers may reduce dissatisfaction and improve retention among customers seeking broader financial solutions.

Retention Challenges Among Customers Aged 40 and Over

Customers aged 40 and above exhibit a sharply increasing churn rate, peaking around age 50, followed by a rapid decline alongside a decreasing customer count. This pattern suggests that engagement and retention strategies for customers over 40 are underperforming.

Further investigation is recommended through improved tracking and analysis of customer complaints and needs within this age segment, as well as identifying the factors that encourage the smaller group of customers above this age threshold to remain with the bank.

Inactive Zero-Balance Accounts

Addressing inactive accounts is also advised, as they represent unrealized value and operational inefficiency. Customers with inactive status and a zero balance account for 14% of the total customer base, comprising 9.5% from France and 4.5% from Spain. No such customers are observed in Germany. Targeted reactivation strategies or account closure policies should be evaluated to optimize the active customer portfolio.